Experimenting that we were so far with small rockets, we neither had the need nor knowledge to set up an integrated test complex. The one we had at Sriharikota was for testing solid propulsion engines of up to 3-tonne thrust. Now, with the mighty 60-tonne Vikas in the making, we needed it.

As for the knowledge, our team in Vernon had acquired the expertise to establish a test complex, though the primary assignment there was putting together the liquid propulsion engine. SEP Vernon had two test stands, one to test the single engine and another, a bigger one, to test a four-engine configuration, which continues to be the first stage of some Arianne vehicles.

We had witnessed several ground firing tests during our stay since 1974, and had closely studied their setting up and functioning. During the tests, as per the protocol, only those designated are allowed in the control room—a small building barely 50 m from the test stand. From there, we would peep through a slit of a window to see with naked eyes the mighty firing of the engine that literally shook the earth while data analysts remained glued to the computers to study the parameters of the firing.

On our return to India, we had the know-how and the ‘know-why’ to set up a test complex, but that was not all. Compared to the solid propulsion engine test facility called STEX in Sriharikota, a liquid engine test complex came with a much higher risk, primarily because it involved handling of chemicals like UDMH (unsymmetrical dimethyl hydrazine), which is used as the fuel for the engine, and N_2O_4 (nitrogen tetroxide) used as the oxidizer. UDMH is carcinogenic; N_2O_4 is equally toxic. In fact, workers were not allowed to deal with these chemicals for a long period—we kept changing them. And we had been wiser by an incident in which a worker fainted after leaning over a much less dangerous red fuming nitric acid (RFNA) tank.

A test complex needed huge space, and there should be no habitation within a radius of at least 3 km. And a test stand is only one part of the facility, which needs space for storing fuel and oxidizer, control rooms, data analysis centres and administrative and residential blocks. This meant hundreds of acres.